Does Better Forecasting Buy Better Budget Pacing?

An Empirical Study of Probabilistic Spend Forecasts for Throttling-Based Daily Pacing on Real Campaign Logs

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Abstract

Demand-side platforms pace daily campaign budgets by throttling participation in auctions. A natural way to pace is to forecast the demand of the remaining day and throttle so that the forecast fits the remaining budget. We ask how much the forecaster and its quantile level matter for this. On 16,468,027 real impressions from 414 campaigns in the Criteo Attribution dataset, we couple rolling-origin probabilistic forecasts (seasonal naive, LightGBM, NHITS and the Temporal Fusion Transformer) with an impression-replay pacing environment. We compare them with no pacing, even pacing and a PID controller tuned on validation days, at three budget levels. Throttling on the median forecast of a neural model captures 3.4–5.4 percentage points more clicks than even pacing and 0.7–2.0 points more than tuned PID, with far smoother participation rates. The hedging quantile matters much more than the forecaster: throttling on the 90th percentile costs 7.7–10.6 points of clicks because it leaves 75.2–82.8% of the budget unspent. TFT and NHITS are practically tied (at most 0.2 points apart). Both beat LightGBM by 0.3–1.0 points and seasonal naive by 1.2–2.1 points. Code, protocol and all results files are released.

Keywords

budget pacing, throttling, real-time bidding, probabilistic forecasting, Temporal Fusion Transformer, empirical study

1 Introduction

Advertisers buy display impressions with daily budgets, and the platform must spend each budget across the day rather than exhaust it in the first busy hours [1, 9]. A widely deployed mechanism is *probabilistic throttling*: the platform takes part in each eligible auction with a probability α_t that it adjusts over the day [1, 2, 13]. How to set α_t is the pacing problem.

Forecast-free controllers, such as even pacing or a PID loop on the spend gap, react to past spend. A forecast-driven policy instead predicts the remaining day's eligible spend and sets α_t so that the predicted spend fits the remaining budget. Modern multi-horizon forecasters, including the Temporal Fusion Transformer (TFT) [10] and NHITS [3], produce such forecasts with calibrated quantiles. A practitioner therefore faces three questions. Does forecasting help over a well-tuned controller? Does a more accurate forecaster translate into better pacing? Should the policy hedge by planning against an upper quantile of demand?

We answer these questions empirically on real campaign logs. This is an applied study: the forecasters and the throttling model are established methods, used as tools. Our contributions are:

(1) A reproducible pipeline that couples hourly rolling-origin probabilistic spend forecasts with an impression-replay throttling environment built from the public Criteo Attribution dataset [4].
(2) A controlled comparison of forecast-free and forecast-driven policies at three budget-tightness levels, with campaign-level bootstrap intervals and multiplicity-corrected paired tests.
(3) Practical guidance: forecasting helps, the hedging quantile dominates the choice of forecaster, and upper-quantile hedging is costly.

The design, metrics and every deviation from them were fixed in a written protocol before the test split was evaluated. Every number reported here is generated from released results files.

2 Related Work

Budget pacing. Lee et al. [9] pace real-time bids with a participation rate adjusted toward a target spend curve. Agarwal et al. [1] describe throttling-based pacing at LinkedIn that uses global supply patterns. Smart pacing [13] learns per-segment pacing rates to also improve performance. Balseiro et al. [2] compare throttling with other budget-management strategies in equilibrium. Optimal bidding under budgets in real-time bidding is studied by Zhang et al. [14]. These works design pacing mechanisms. We instead hold the mechanism fixed and vary the quality and hedging level of the demand forecast inside it.

Forecasting. TFT [10] combines recurrent encoding, attention and variable selection for multi-horizon quantile forecasts. NHITS [3] uses hierarchical interpolation with multi-rate sampling. Gradient-boosted trees such as LightGBM [7] are strong tabular baselines. We use the `neuralforecast` implementations [11] rather than custom code, and we evaluate accuracy with MASE [6] and the pinball loss [8].

3 Pacing Setup

Episodes and budgets. An episode is one campaign-day of 24 hours. Campaign c receives the daily budget $B = \rho\,\bar{D}_c$, where $\bar{D}_c$ is its mean daily spend over the training days and $\rho \in \{0.3, 0.5, 0.7\}$ sets the budget tightness. In 94.3%, 86.5% and 70.8% of test episodes, respectively, the day's logged demand exceeds B, so the budget binds.

Throttling environment. At the start of hour h the policy sets $\alpha_h \in [0, 1]$. Each logged impression of that hour is kept independently with probability α_h. Kept impressions are served in log order, with their logged cost, click and conversion, until the

next impression would exceed the remaining budget. The uniform draws are fixed per replay seed, campaign and hour, so all policies face identical randomness. We assume that throttling does not change the price or value of the kept impressions. This is the standard throttling model, and we make no claim about bid-shading interactions.

Policies. Let S_h be the spend before hour h and $R_h = B - S_h$.

- *No pacing*: $\alpha_h = 1$ until the budget is exhausted.
- *Even pacing*: starts at $\alpha_0 = \min(1, B/\bar{D}_c)$. It then corrects multiplicatively toward an equal spend of $R_h/(24-h)$ per remaining hour.
- *PID*: $\alpha_h = \alpha_0 + K_p e_h + K_i \sum_{j \le h} e_j + K_d(e_h - e_{h-1})$ on the normalized gap $e_h = (Bh/24 - S_h)/B$. The gains are grid-searched on validation days separately for each ρ.
- *Forecast throttle*: at every hour, a fresh forecast $\hat{y}_h^{(q)}, \ldots, \hat{y}_{23}^{(q)}$ of the q-quantile hourly demand gives

$$\alpha_h = \min\Big(1,\ R_h \Big/ \textstyle\sum_{j=h}^{23} \hat{y}_j^{(q)}\Big), \qquad q \in \{0.5, 0.7, 0.9\}. \tag{1}$$

- *Hindsight constant* (reference): $\alpha = \min(1, B/D)$ with the day's true demand D. It is the denominator of our primary metric. It is not an upper bound, because a time-varying rate can capture more clicks than a constant one.

Raising q in Eq. (1) inflates the planned demand and lowers α_h. This is the "hedging" knob: it trades spend for protection against early depletion.

4 Forecasting Models

The target is hourly campaign spend. Forecasts are issued at every hour of the test period for the next 24 hours, using only earlier hours. All models output the quantiles $\{0.1, 0.5, 0.7, 0.9\}$.

- *Seasonal naive*: empirical quantiles of the same hour on the previous seven days.
- *LightGBM*: a direct multi-horizon model with the forecast step as a feature. It is trained with the quantile objective, one model per quantile. Features are spend lags, same-hour seasonal lags, rolling statistics, calendar phase and static campaign statistics, with the target normalized by the campaign's training mean.
- *TFT* and *NHITS*: `neuralforecast` models with a 168-hour input window, multi-quantile loss, robust scaling, historical impressions and clicks, calendar phase as known future inputs, and static campaign statistics.

Models are trained on the training days and early-stopped on the validation days. Hyperparameters were fixed before evaluation and not tuned on test data. The neural models are trained with 3 seeds each.

5 Experimental Protocol

Data. The Criteo Attribution dataset [4] contains 16,468,027 impressions from 675 campaigns over 31 days, with the paid cost (transformed by Criteo), click and attributed conversion of each impression. The first 21 days are for training, the next 4 for validation and the following 5 for testing. A campaign is included if it has spend in at least 90% of training hours. The inclusion rule uses training data only. It keeps 414 campaigns, which account for 91.7% of all spend, and gives 2,070 test episodes, each replayed with 10 seeds.

Metrics. Forecast accuracy is measured with MASE (scaled by each campaign's in-sample lag-24 naive error), the scaled pinball loss sPL_q, the 80% interval coverage and the quantile-crossing rate. Pacing is measured by the ratio of clicks (and of attributed conversions) to the hindsight reference, pooled over episodes; budget utilization (BUR); the early-depletion rate (budget exhausted before hour 20); and the pacing volatility index (PVI), the RMS change of α_h between consecutive hours.

Statistics. We report 95% confidence intervals from a bootstrap over campaigns (2,000 resamples). Pre-specified paired comparisons use Wilcoxon signed-rank tests [12] on per-campaign click ratios, with a Holm correction [5] over all 126 comparisons.

Protocol. The study protocol, including a deviation log, was written before the test split was evaluated. All logged deviations are released, including two widenings of the PID grid (on validation data) and intermittent GPU faults during neural training (crashed runs produced no output and were re-run).

6 Results

6.1 Forecast accuracy

All four forecasters beat the in-sample seasonal-naive scale (Table 2). TFT (MASE 0.787) and NHITS (0.794) are close and both clearly ahead of LightGBM (0.853) and seasonal naive (0.917). The neural models' advantage is largest at short steps (Figure 2). All intervals are too narrow: the nominal 80% interval covers only 76.9% of outcomes for TFT and 60.6% for seasonal naive.

6.2 Pacing outcomes

Table 1 and Figure 1 summarize pacing. Throttling on the median forecast of a neural model is the best policy at every budget level. It captures 3.4–5.4 points more clicks than even pacing and 0.7–2.0 points more than tuned PID. All 12 of 12 paired comparisons against these two baselines are significant after Holm correction (largest adjusted p = 0.00052). It is also far smoother: its PVI of 0.08–0.12 compares with 0.43–0.60 for PID, and it depletes early in only 3.4–6.1% of episodes, against 12.8–31.9% for PID.

The forecaster matters little. TFT and NHITS differ by at most 0.2 points of clicks; only 1 of 3 of their paired comparisons is significant. The neural models beat LightGBM by 0.3–1.0 points (significant in 6 of 6 comparisons) and seasonal naive by 1.2–2.1 points. Even seasonal naive at q=0.5 is competitive with tuned PID.

The hedging quantile matters a lot. Moving from q=0.5 to q=0.9 costs 7.7–10.6 points of clicks, for every forecaster and budget level. All 24 comparisons of q=0.9 policies against even pacing and PID favor the baselines. Upper-quantile planning overstates the remaining demand, so the policy throttles too hard and leaves budget unspent.

6.3 The hedging trade-off

Figure 3 shows what hedging buys. At q=0.9, early depletion almost disappears (0.1–0.7% of episodes), but budget utilization falls to 75.2–82.8%. No pacing sits at the opposite corner: it depletes early in 59.1–92.1% of episodes. The median forecast with a neural model gives the best combination of the two.

Table 1: Pacing outcomes on the test days. Clicks are relative to the hindsight constant-rate reference, pooled over campaign-days and replay seeds; the best non-reference value per budget level is in bold. The right block reports $\rho = 0.5$. Bootstrap CIs for all entries are in the released `pacing_summary.csv`.

	Clicks / reference			At $\rho = 0.5$			
Policy	ρ=0.3	ρ=0.5	ρ=0.7	Conv./ref.	BUR (%)	Early depl. (%)	PVI
Hindsight constant (ref.)	1.000	1.000	1.000	1.000	90.6	9.5	0.000
No pacing	0.985	0.979	0.988	1.018	94.4	80.1	0.000
Even pacing	0.993	0.960	0.942	0.966	90.2	6.4	0.314
PID (tuned)	1.009	0.990	0.988	1.004	92.2	31.9	0.577
Seasonal naive, q=0.5	1.007	0.989	0.983	0.988	91.3	6.4	0.083
LightGBM, q=0.5	1.019	0.997	0.992	0.998	92.4	5.0	0.094
NHITS, q=0.5	**1.028**	**1.007**	0.995	1.005	92.5	4.3	0.118
TFT, q=0.5	1.027	1.007	**0.996**	1.006	92.7	5.5	0.097
Seasonal naive, q=0.7	0.984	0.965	0.959	0.957	87.7	2.4	0.075
LightGBM, q=0.7	0.995	0.972	0.965	0.964	88.7	1.6	0.082
NHITS, q=0.7	1.011	0.985	0.971	0.976	89.1	1.1	0.101
TFT, q=0.7	1.010	0.985	0.974	0.977	89.4	1.6	0.085
Seasonal naive, q=0.9	0.930	0.910	0.904	0.897	79.6	0.6	0.070
LightGBM, q=0.9	0.914	0.892	0.887	0.879	78.2	0.5	0.071
NHITS, q=0.9	0.950	0.919	0.904	0.904	79.6	0.2	0.087
TFT, q=0.9	0.945	0.916	0.904	0.905	79.6	0.3	0.074

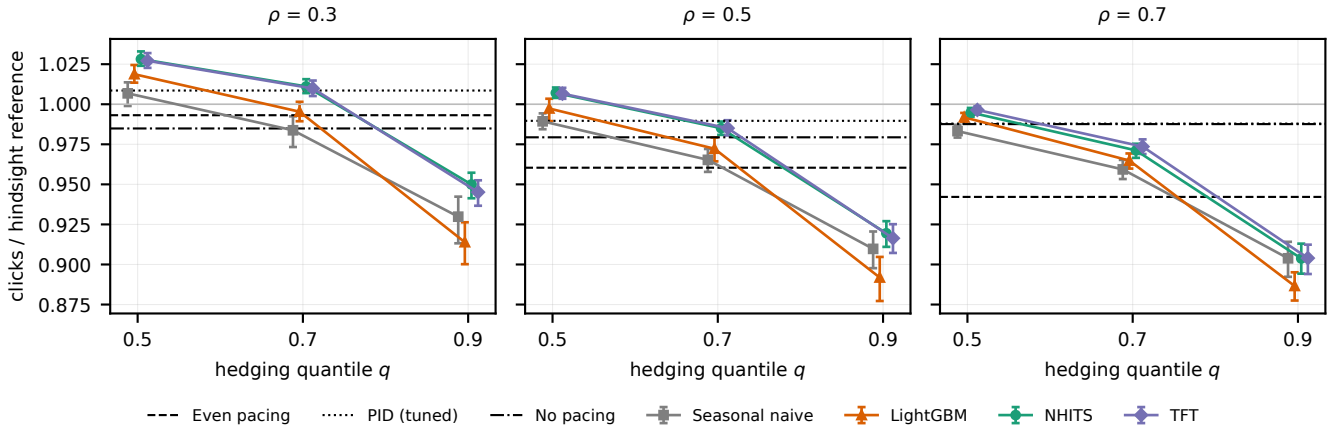


Figure 1: Clicks relative to the hindsight reference as a function of the hedging quantile, with 95% campaign-bootstrap CIs. Horizontal lines are the forecast-free policies.

Table 2: Forecast accuracy on the test origins, averaged over campaigns; ± is the standard deviation over seeds. Time is the wall-clock time to fit and forecast (neural models on one RTX 3070 Ti, LightGBM on an Apple M4 CPU).

Forecaster	MASE	$\text{MASE}_{h=1}$	$\text{MASE}_{h=24}$	$\text{sPL}_{0.9}$	Cov. 80%	Time (s)
Seasonal naive	0.917	0.920	0.908	0.194	60.6	0
LightGBM	0.853	0.810	0.871	0.185	71.7	76
NHITS	0.794 ± 0.014	0.719	0.813	0.158	75.8	40
TFT	0.787 ± 0.004	0.729	0.814	0.161	76.9	205

Conversions behave differently from clicks. No pacing captures 1.4–3.2 points *more* attributed conversions than the reference, while capturing fewer clicks. One possible explanation, which we have not tested, is that traffic early in the (dataset-relative) day converts better than it clicks. Either way, an operator who optimizes for conversions should not assume that click-optimal pacing is also conversion-optimal.

7 Discussion

Practical guidance. (i) Forecast-driven throttling is worth deploying over even pacing and over a tuned PID. (ii) Plan against the median: the extra protection from upper quantiles is small and expensive. If early depletion must be avoided, q=0.7 is a more balanced choice than q=0.9. (iii) Forecast accuracy gains shrink when they reach pacing. A plausible reason, not tested here, is that the

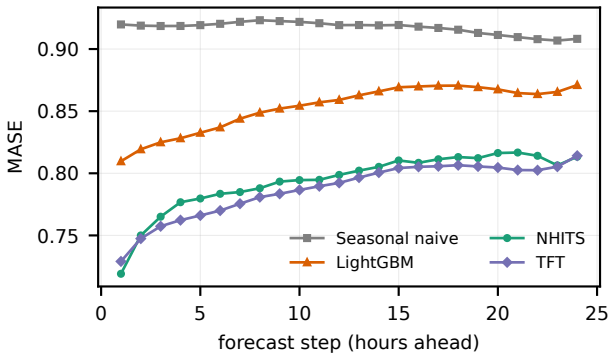


Figure 2: MASE by forecast step, averaged over campaigns and seeds.

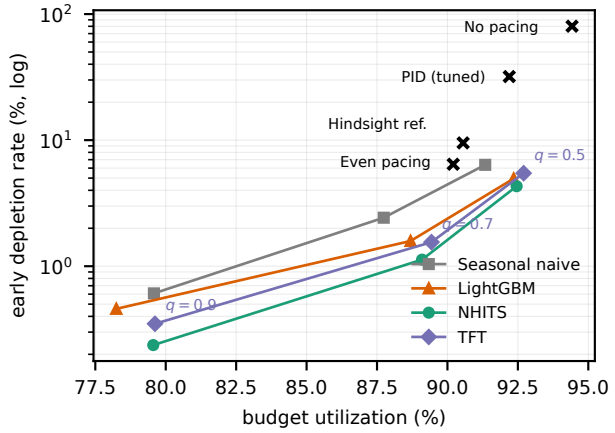


Figure 3: Delivery trade-off at $\rho = 0.5$: budget utilization against the early-depletion rate (log scale). Lines connect $q = 0.5, 0.7, 0.9$ of each forecaster.

throttle in Eq. (1) depends only on the *sum* of the remaining forecasts, where hourly errors can partly cancel. A cheaper model such as NHITS (40 s versus 205 s for TFT) loses nothing here.

Limitations. The costs in the dataset are transformed by Criteo. The throttling model ignores competitive responses and price effects. The test period covers five days of one dataset. The day boundary is relative to the dataset start, not local midnight. The tuned PID gains lie on the edge of the search grid (for example K_p=0.03125, K_i=1, K_d=32 at ρ=0.5), although the validation objective is flat near the optimum. With hundreds of campaigns, small differences become statistically significant, so we emphasize effect sizes.

8 Conclusion

On real campaign logs, forecast-driven throttling improves daily budget pacing over even pacing and tuned PID. The main lever is the hedging quantile, not the forecasting architecture: the median forecast works best, and upper-quantile hedging is costly. Modern neural forecasters are the most accurate and give the best pacing, but their advantage over each other is negligible. Future work should test these findings with bid-landscape-aware simulators and with conversion-oriented objectives.

References

[1] Deepak Agarwal, Souvik Ghosh, Kai Wei, and Siyu You. 2014. Budget Pacing for Targeted Online Advertisements at LinkedIn. In *Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. doi:10.1145/2623330.2623366

[2] Santiago Balseiro, Anthony Kim, Mohammad Mahdian, and Vahab Mirrokni. 2017. Budget Management Strategies in Repeated Auctions. In *Proceedings of the 26th International Conference on World Wide Web (WWW)*. 15–23.

[3] Cristian Challu, Kin G. Olivares, Boris N. Oreshkin, Federico Garza Ramirez, Max Mergenthaler-Canseco, and Artur Dubrawski. 2023. NHITS: Neural Hierarchical Interpolation for Time Series Forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. doi:10.1609/aaai.v37i6.25854

[4] Eustache Diemert, Julien Meynet, Pierre Galland, and Damien Lefortier. 2017. Attribution Modeling Increases Efficiency of Bidding in Display Advertising. In *Proceedings of the AdKDD and TargetAd Workshop, KDD*. arXiv:1707.06409.

[5] Sture Holm. 1979. A Simple Sequentially Rejective Multiple Test Procedure. *Scandinavian Journal of Statistics* 6, 2 (1979), 65–70.

[6] Rob J. Hyndman and Anne B. Koehler. 2006. Another Look at Measures of Forecast Accuracy. *International Journal of Forecasting* 22, 4 (2006), 679–688.

[7] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. 2017. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. In *Advances in Neural Information Processing Systems 30 (NIPS)*.

[8] Roger Koenker and Gilbert Bassett. 1978. Regression Quantiles. *Econometrica* 46, 1 (1978), 33–50.

[9] Kuang-Chih Lee, Ali Jalali, and Ali Dasdan. 2013. Real Time Bid Optimization with Smooth Budget Delivery in Online Advertising. In *Proceedings of the Seventh International Workshop on Data Mining for Online Advertising (ADKDD)*.

[10] Bryan Lim, Sercan Ö. Arık, Nicolas Loeff, and Tomas Pfister. 2021. Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting. *International Journal of Forecasting* 37, 4 (2021), 1748–1764.

[11] Kin G. Olivares, Cristian Challú, Federico Garza, Max Mergenthaler Canseco, and Artur Dubrawski. 2022. NeuralForecast: User Friendly State-of-the-Art Neural Forecasting Models. PyCon, Salt Lake City, Utah, US. https://github.com/Nixtla/neuralforecast.

[12] Frank Wilcoxon. 1945. Individual Comparisons by Ranking Methods. *Biometrics Bulletin* 1, 6 (1945), 80–83.

[13] Jian Xu, Kuang-Chih Lee, Wentong Li, Hang Qi, and Quan Lu. 2015. Smart Pacing for Effective Online Ad Campaign Optimization. In *Proceedings of the 21st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*.

[14] Weinan Zhang, Shuai Yuan, and Jun Wang. 2014. Optimal Real-Time Bidding for Display Advertising. In *Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*.